

Regularization & Cost Function - 5 Numerical Questions

Q1. L2 Regularized Cost Function

Given $m = 5$, prediction errors are 2, -1, 3, -2, and 1. The model has weights $\theta_1 = 2$ and $\theta_2 = -3$, with $\lambda = 0.4$.

Calculate:

1. The original cost function $J(\theta)$ using $J(\theta) = (1/2m) \sum (h_{\theta}(x^{(i)}) - y^{(i)})^2$.
2. The L2 regularization term $\lambda/(2m) \sum \theta_j^2$.
3. The total regularized cost.

Q2. L1 Regularized Cost Function

Given $m = 4$, prediction errors are 1, -2, 2, and -1. The weights are $\theta_1 = 3$ and $\theta_2 = -4$, with $\lambda = 0.5$.

Calculate:

1. The original cost function $J(\theta)$.
2. The L1 regularization term $\lambda/(2m) \sum |\theta_j|$.
3. The total regularized cost.

Q3. Compare Regularized and Unregularized Cost

For $m = 5$, the squared prediction errors are 4, 1, 9, 4, and 1. The weights are $\theta_1 = 2$, $\theta_2 = -2$, and $\theta_3 = 1$. Let $\lambda = 0.5$.

Calculate:

1. The unregularized cost $J(\theta)$.
2. The L2 regularization penalty using $\lambda/(2m) \sum \theta_j^2$.
3. The regularized cost.
4. The increase in cost due to regularization.

Q4. Ridge Cost Function

A linear regression model has $m = 10$ observations. The sum of squared errors is 80. The model weights are $\theta_1 = 3$, $\theta_2 = -4$, and $\theta_3 = 2$. Use $\lambda = 0.2$.

Calculate the Ridge cost using:

$$J_{\text{Ridge}}(\theta) = \text{SSE}/(2m) + \lambda/(2m) \sum \theta_j^2.$$

Q5. Elastic Net Cost Function

Given $m = 6$, the sum of squared errors is 48. The weights are $\theta_1 = 2$ and $\theta_2 = -3$. Let $\lambda = 0.6$ and $\alpha = 0.7$.

Calculate the Elastic Net cost using:

$$J(\theta) = \text{SSE}/(2m) + \lambda/(2m) [\alpha \sum |\theta_j| + (1 - \alpha) \sum \theta_j^2].$$